

Programming Fundamentals

Assessment 1

March 26, 2026

Write a program to calculate the water bill of a house when consumed units entered according to the following criteria.

Units Range	Cost per unit
0 - 30	5
31 - 60	10
61 - 100	15
More than 100	50
Government tax of 500/= must be added to all the bills.	

activity.py

```
1  # Write a program to calculate a water bill of a house when consumed units entered according to follolwing criteria,
2  #
3  # Unit Range      |      Cost per unit
4  # 0-30           |      5
5  # 31-60          |      10
6  # 61-100         |      15
7  # more than 100  |      50
8  ⚡
9
10 def calculate_water_bill(units_count: int) -> int:
11     if units_count <= 30:
12         return units_count * 5 + 500
13     elif units_count <= 60:
14         return 30 * 5 + (units_count - 30) * 10 + 500
15     elif units_count <= 100:
16         return 30 * 5 + 30 * 10 + (units_count - 60) * 15 + 500
17     else:
18         return 30 * 5 + 30 * 10 + 40 * 15 + (units_count - 100) * 50 + 500
19
20
21 def main() -> None:
22     while True:
23         try:
24             units_count: int = int(input("Enter water units count: "))
25             break
26         except ValueError:
27             print("Invalid value. Retry.")
28
29     if units_count >= 0:
30         bill: int = calculate_water_bill(units_count)
31         print(f"Your water bill is: {bill}")
32     else:
33         print("Units count cannot be negative. Restart")
34
35
> 36 if __name__ == "__main__":
37     main()
38
```

pf-4 — pwsh.exe

```
pf-4> uv run activity.py
Enter water units count: something
Invalid value. Retry.
Enter water units count: 25
Your water bill is: 625
pf-4> uv run activity.py
Enter water units count: 35
Your water bill is: 700
pf-4> uv run activity.py
Enter water units count: 65
Your water bill is: 1025
pf-4> uv run activity.py
Enter water units count: 105
Your water bill is: 1800
pf-4> 
```